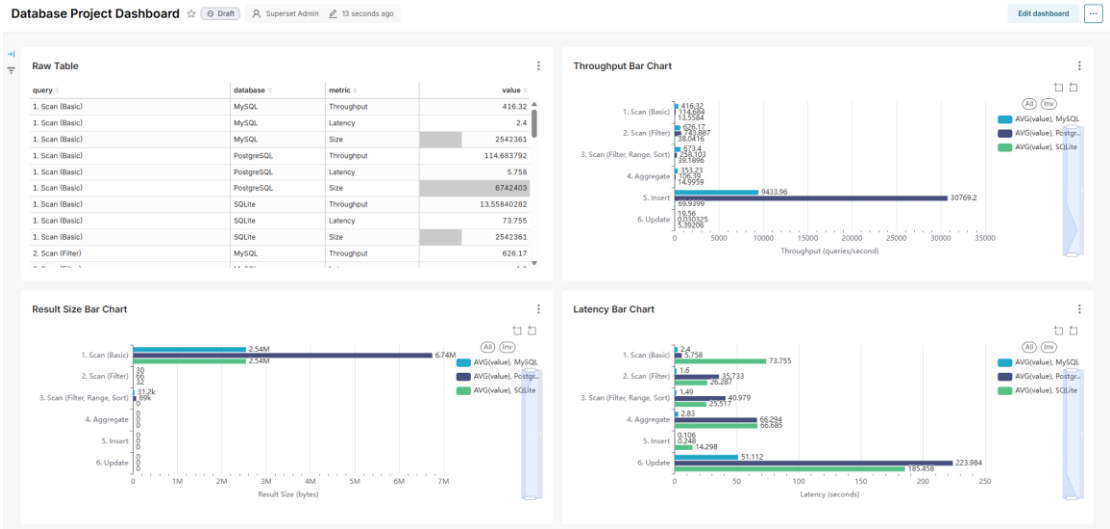
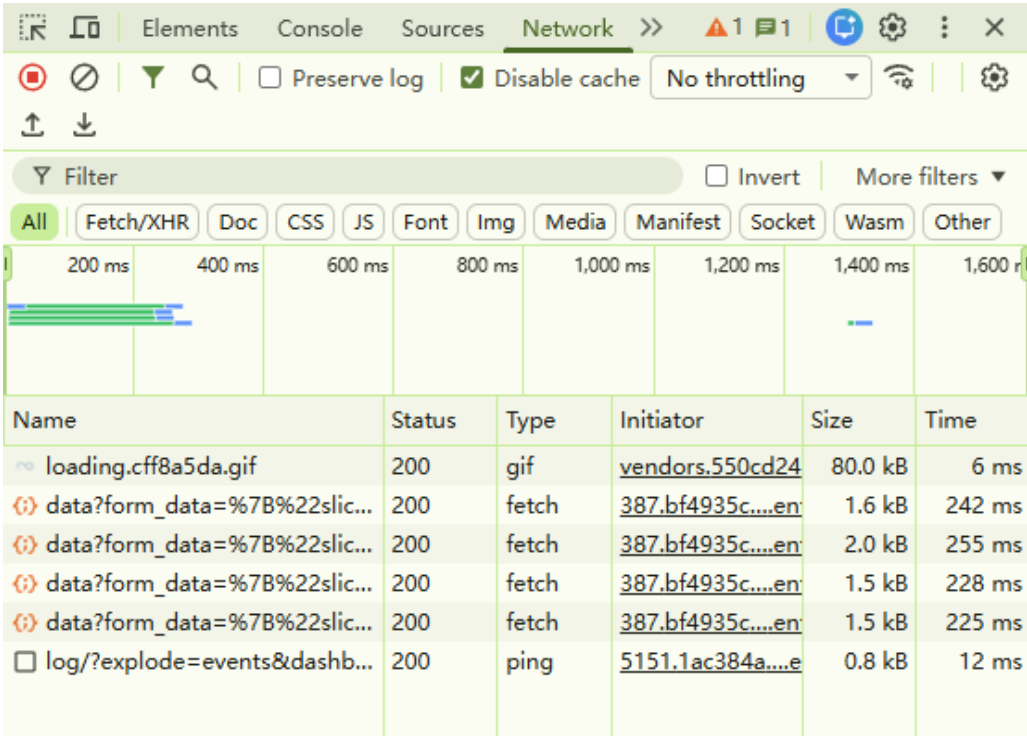


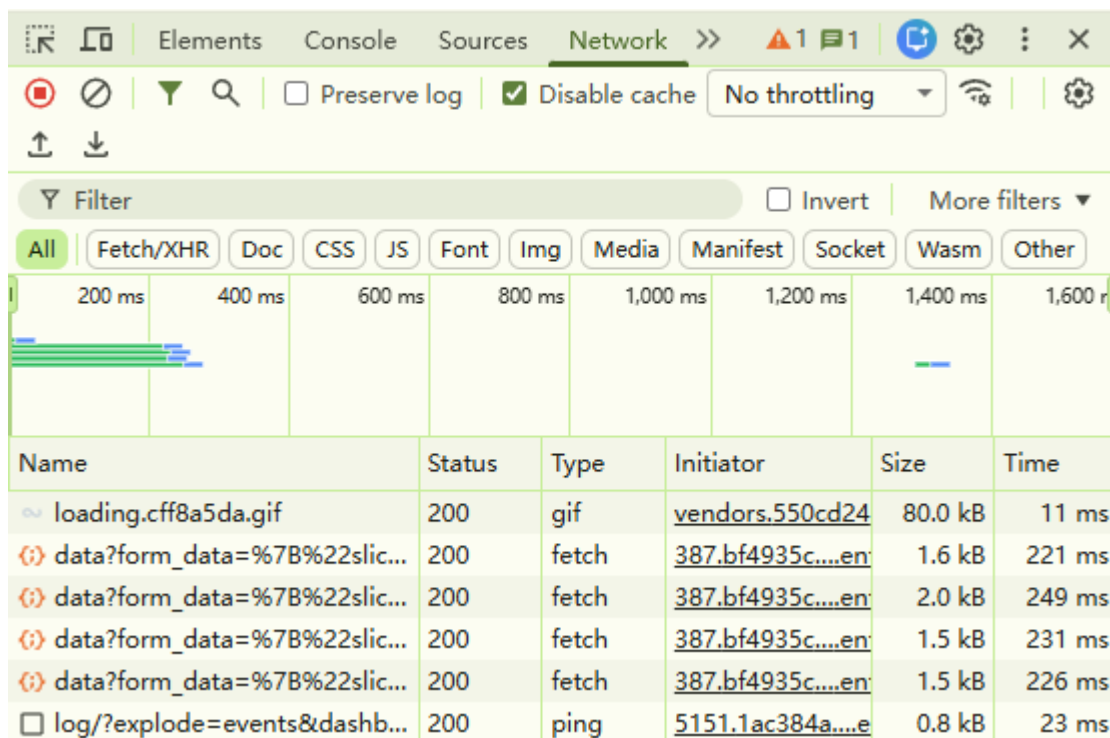
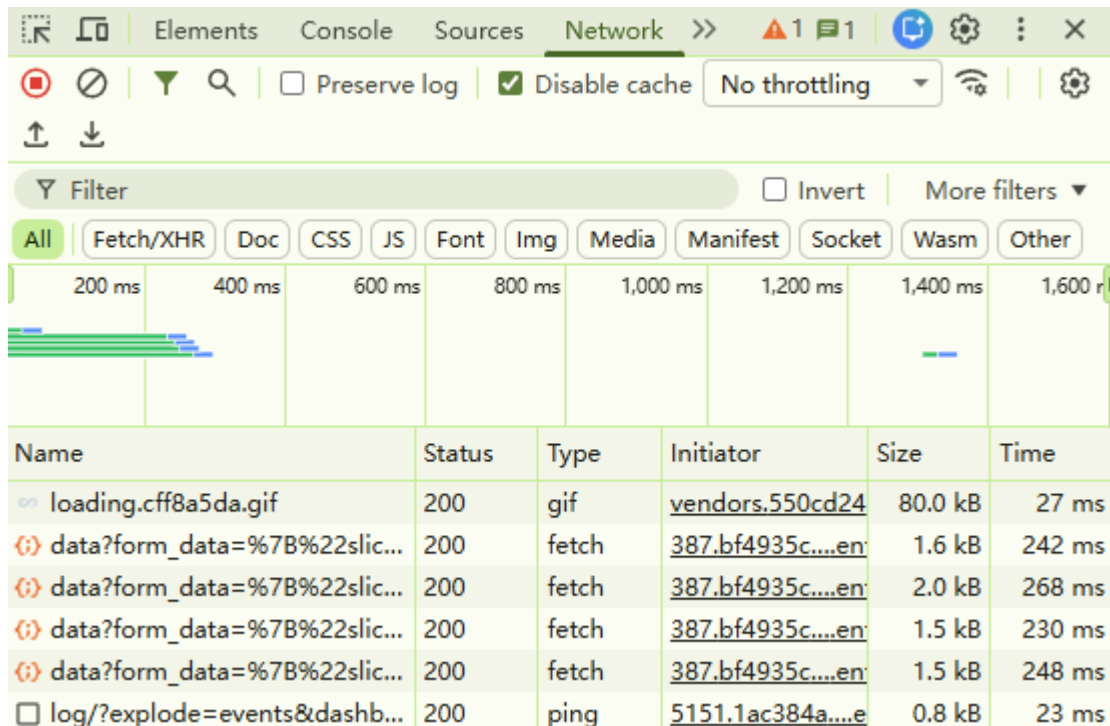
We utilize the Superset platform to visualize the data processing capabilities of three databases: MySQL, PostgreSQL, and SQLite. In this project, we present the initial data in tabular form, while comparing three measurement parameters—Throughput, Result size, and Latency via bar charts.



Next, we will measure the relevant performance metrics of the Superset platform. Due to the lack of built-in tools, we manually capture the time using the browser's Developer Tools.

First, we test the Dashboard load time. Manually refresh the Dashboard and use the statistics at the bottom of the Network tab(with caching disabled). We can measure the specific refresh time for each chart and the total time required to refresh the entire dashboard. Perform this measurement three times and take the average.





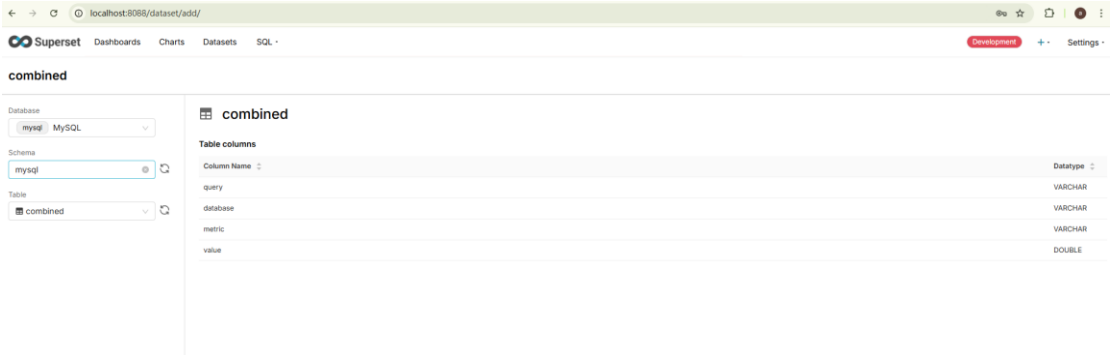
We can calculate the average dashboard load time as follows: $(956 + 1015 + 938) / 3 = 969.67\text{ms}$

Next, we test interactive latency, mainly includes:

- Load dataset into Superset
- Connect Superset to three databases

Load dataset into Superset

We uploaded the pre-configured tables from our database to Superset.



Through backend log records, we can observe the entire process and duration of loading the dataset into Superset.

	Superset Admin	ExploreFormDataRestApi.put	2025-11-22 23:22:58.456153
	Superset Admin	ChartDataRestApi.data	2025-11-22 23:22:57.579046
	Superset Admin	_get_data_response	2025-11-22 23:22:57.574583
	Superset Admin	ChartDataRestApi.json_dumps	2025-11-22 23:22:57.569548
	Superset Admin	QueryObject.post_processing	2025-11-22 23:22:57.559983
	Superset Admin	load_into_dataframe	2025-11-22 23:22:57.548688
	Superset Admin	fetch_rows	2025-11-22 23:22:57.541714
	Superset Admin	execute_sql	2025-11-22 23:22:57.534161

For example:

Show Log	
User	Superset Admin
Action	execute_sql
Dashboard Id	None
Slice Id	None
JSON	{ "path": "/api/v1/chart/data", "datasource": { "id": 35, "type": "table", "force": false, "queries": [{ "filters": [], "extras": { "time_grain_sqla": "PID", "having": "", "where": "" }, "applied_time_extras": {}, "columns": ["query", "database", "metric", "value"], "orderby": [], "annotation_layers": [], "row_limit": 1000, "series_limit": 0, "order_desc": true, "url_params": { "datasource_id": "35", "datasource_type": "table", "custom_params": {}, "custom_form_data": {}, "post_processing": [], "time_offsets": [] }, "form_data": { "filters": [], "extras": { "time_grain_sqla": "PID", "having": "", "where": "" }, "applied_time_extras": {}, "columns": ["query", "database", "metric", "value"], "orderby": [], "annotation_layers": [], "row_limit": 1000, "series_limit": 0, "order_desc": true, "url_params": { "datasource_id": "35", "datasource_type": "table", "custom_params": {}, "custom_form_data": {}, "post_processing": [], "time_offsets": [] }, "result_format": "json", "result_type": "full", "object_ref": "superset.models.core" } }] }
dttn	2025-11-22 23:22:57.534161
Duration Ms	10
Referrer	http://localhost:8088/explore/?form_data_key=Fh4SZ09etPg&datasource_type=table&datasource_id=35

Show Log

User	Superset Admin
Action	load_into_dataframe
Dashboard Id	None
Slice Id	None
JSON	<pre>{ "path": "/api/v1/chart/data", "datasource": { "id": 35, "type": "table", "force": false, "queries": [{ "filters": [], "extras": { "time_grain_sqla": "P1D", "having": "", "where": "" }, "applied_time_extras": {}, "columns": ["query", "database", "metric", "value"], "orderby": [], "annotation_layers": [], "row_limit": 1000, "series_limit": 0, "order_desc": true, "url_params": { "datasource_id": "35", "datasource_type": "table", "custom_params": {}, "custom_form_data": {}, "post_processing": [], "time_offsets": [] }, "form_data": { "filters": [], "extras": { "time_grain_sqla": "P1D", "having": "", "where": "" }, "applied_time_extras": {}, "columns": ["query", "database", "metric", "value"], "orderby": [], "annotation_layers": [], "row_limit": 1000, "series_limit": 0, "order_desc": true, "url_params": { "datasource_id": "35", "datasource_type": "table", "custom_params": {}, "custom_form_data": {}, "post_processing": [], "time_offsets": [] }, "result_format": "json", "result_type": "full", "object_ref": "Database.load_into_dataframe" } }] }, "dtm": "2025-11-22 23:22:57.548688", "duration_ms": 2, "referrer": "http://localhost:8088/explore/?form_data_key=Fh4SZ09etPg&datasource_type=table&datasource_id=35" }</pre>

Adding up the total time for the entire process: $10 + 1 + 2 + 0 + 0 + 82 + 130 + 17 = 242$ ms

Then we test **connecting Superset to three databases**

Postgresql:

The entire process:

Q	Superset Admin	DatabaseRestApi.available	2025-11-22 23:35:02.975538
Q	Superset Admin	DatabaseRestApi.get_list	2025-11-22 23:35:02.955938
Q	Superset Admin	DatabaseRestApi.get_list	2025-11-22 23:35:02.948122
Q	Superset Admin	DatabaseRestApi.put	2025-11-22 23:35:02.844991
Q	Superset Admin	DatabaseRestApi.validate_parameters	2025-11-22 23:35:02.338704
Q	Superset Admin	DatabaseRestApi.available	2025-11-22 23:35:01.225986
Q	Superset Admin	DatabaseRestApi.get_list	2025-11-22 23:35:01.209322

Partial Detailed Records:

Show Log

User	Superset Admin
Action	DatabaseRestApi.validate_parameters
Dashboard Id	None
Slice Id	None
JSON	<pre>{ "path": "/api/v1/database/validate_parameters/", "allow_ctas": false, "allow_cvas": false, "allow_dml": false, "allow_file_upload": true, "allow_run_async": false, "backend": "postgresql", "cache_timeout": null, "configuration_method": "dynamic_form", "database_name": "PostgreSQL", "driver": "psycopg2", "engine_information": { "disable_ssh_tunneling": false, "supports_dynamic_catalog": true, "supports_file_upload": true, "supports_output2": false, "expose_in_sdlab": true, "extra": {} }, "allows_virtual_table_explore": true, "schemas_allowed_for_file_upload": ["public"], "force_ctas_schema": null, "id": 2, "impersonate_user": false, "is_managed_externally": false, "masked_encrypted_extra": {}, "parameters": { "database": "postgres", "encryption": false, "host": "host.docker.internal", "password": "XXXXXXXXXX", "port": 5432, "query": {}, "username": "postgres", "parameters_schema": { "properties": { "database": { "description": "Database name", "type": "string", "encryption": { "description": "Use an encrypted connection to the database", "type": "boolean" }, "host": { "description": "Hostname or IP address", "type": "string", "password": { "description": "Password", "nullable": true, "type": "string" }, "port": { "description": "Database port", "maximum": 65536, "minimum": 0, "type": "integer", "query": { "additionalProperties": {} }, "description": "Additional parameters", "type": "object", "ssh": { "description": "Use an ssh tunnel connection to the database", "type": "boolean", "username": { "description": "Username", "nullable": true, "type": "string" } }, "required": { "database", "host", "port", "username" }, "type": "object", "server_cert": null, "sqlalchemy_uri": "postgresql+psycopg2://postgres:XXXXXXXXXX@host.docker.internal:5432/postgres", "uid": "7fa36a1f-3c68-42f0-bfd5-7e58cb2075ae", "engine": "postgresql", "catalog": {}, "query_input": {}, "object_ref": "DatabaseRestApi.validate_parameters" } } } } } }, "dtm": "2025-11-22 23:35:02.338704", "duration_ms": 155, "referrer": "http://localhost:8088/databaseview/list/?pageIndex=0&sortColumn=changed_on_delta_humanized&sortOrder=desc" }</pre>

←

Adding up the total time for the entire process: $28 + 53 + 155 + 485 + 32 + 34 + 63 = 850$ ms

MYSQL:

The entire process:

	User	Action	dtm
	Superset Admin	DatabaseRestApi.available	2025-11-22 23:39:31.856609
	Superset Admin	DatabaseRestApi.get_list	2025-11-22 23:39:31.832312
	Superset Admin	DatabaseRestApi.get_list	2025-11-22 23:39:31.827635
	Superset Admin	DatabaseRestApi.put	2025-11-22 23:39:31.720685
	Superset Admin	DatabaseRestApi.validate_parameters	2025-11-22 23:39:31.612651
	Superset Admin	DatabaseRestApi.available	2025-11-22 23:39:30.204405
	Superset Admin	DatabaseRestApi.get_list	2025-11-22 23:39:30.184886

Partial Detailed Records:

Show Log

User	Superset Admin
Action	DatabaseRestApi.put
Dashboard Id	None
Slice Id	None
JSON	{ "path": "/api/v1/database/5", "allow_ctas": false, "allow_cvas": false, "allow_dml": false, "allow_file_upload": false, "allow_run_async": false, "backend": "mysql", "cache_timeout": null, "configuration_method": "dynamic_form", "database_name": "MySQL", "engine_information": { "disable_ssh_tunneling": false, "supports_dynamic_catalog": false, "supports_file_upload": true, "supports_oauth2": false }, "expose_in_sqllab": true, "extra": "{ \"allows_virtual_table_explore\": true }", "force_ctas_schema": null, "id": 5, "impersonate_user": false, "is_managed_externally": false, "masked_encrypted_extra": {}, "parameters_schema": { "properties": { "database": { "description": "Database name", "type": "string", "encryption": { "description": "Use an encrypted connection to the database", "type": "boolean", "host": { "description": "Hostname or IP address", "type": "string", "password": { "description": "Password", "nullable": true, "type": "string", "port": { "description": "Database port", "maximum": 65536, "minimum": 0, "type": "integer", "query": { "additionalProperties": {}, "description": "Additional parameters", "type": "object", "ssh": { "description": "Use an ssh tunnel connection to the database", "type": "boolean", "username": { "description": "Username", "nullable": true, "type": "string", "required": ["database", "host", "port", "username",], "type": "object", "server_cert": null, "sqlalchemy_uri": "mysql+mysqldb://root:XXXXXXXXX@host.docker.internal:3306/mysql", "uuid": "f8ebe3e8-6b98-4588-b52c-14a0472bde4b", "catalog": [], "query_input": "", "url_rule": "/api/v1/database/<int:pk>", "object_ref": "DatabaseRestApi.put", "pk": 5
dtm	2025-11-22 23:39:31.720685
Duration Ms	79
Referrer	http://localhost:8088/databaseview/list/?pageIndex=0&sortColumn=changed_on_delta_humanized&sortOrder=desc

Adding up the total time for the entire process: 38+61+55+79+29+25+54=341ms

Sqlite:

The entire process:

	User	Action	dtm
	Superset Admin	DatabaseRestApi.available	2025-11-22 23:44:07.518049
	Superset Admin	DatabaseRestApi.get_list	2025-11-22 23:44:07.496864
	Superset Admin	DatabaseRestApi.get_list	2025-11-22 23:44:07.495900
	Superset Admin	DatabaseRestApi.put	2025-11-22 23:44:07.409902
	Superset Admin	DatabaseRestApi.available	2025-11-22 23:44:05.627585
	Superset Admin	DatabaseRestApi.get_list	2025-11-22 23:44:05.601917

Partial Detailed Records:

Show Log

User	Superset Admin
Action	DatabaseRestApi.put
Dashboard Id	None
Slice Id	None
JSON	{ "path": "/api/v1/database/4", "allow_ctas": false, "allow_cvas": false, "allow_dml": false, "allow_file_upload": true, "allow_run_async": false, "backend": "sqlite", "cache_timeout": null, "configuration_method": "sqlalchemy_form", "database_name": "SQLite1", "engine_information": { "disable_ssh_tunneling": true, "supports_dynamic_catalog": false, "supports_file_upload": true, "supports_oauth2": false }, "expose_in_sqllab": true, "extra": { "allows_virtual_table_explore": true }, "force_ctas_schema": null, "id": 4, "impersonate_user": false, "is_managed_externally": false, "masked_encrypted_extra": {}, "parameters_schema": {}, "server_cert": null, "sqlalchemy_url": "sqlite:///app/localdbs/main.db", "uuid": "fa258ccc-0fc5-43ed-9802-1b3398299d57", "query_input": "", "url_rule": "/api/v1/database/<int:pk>", "object_ref": "DatabaseRestApi.put", "pk": 4 }
dtm	2025-11-22 23:44:07.409902
Duration Ms	51
Referrer	http://localhost:8088/databaseview/list/?pageIndex=0&sortColumn=changed_on_delta_humanized&sortOrder=desc

Adding up the total time for the entire process: 34+51+51+19+28+44=227ms

The test results clearly reveal the fundamental differences in architecture design and connection overhead among the three databases: SQLite (227ms) benefits from its embedded architecture, bypassing network protocol stacks and involving only local file I/O, making it the fastest to connect; MySQL (341ms) employs an efficient multi-threaded model to handle client connections, incurring moderate network handshake overhead; whereas PostgreSQL (850ms), traditionally built on a multi-process architecture, includes more complex initialization and security handshake procedures during connection establishment, resulting in the highest initial connection latency.